19.28 PROPERTIES OF EIGENVECTORS

1. The eigenvector X of a matrix A is not unique.

2. If $\lambda_1, \lambda_2, \ldots \lambda_n$ be distinct eigenvalues of an $n \times n$ matrix then corresponding eigen vectors $X_1, X_2, \ldots X_n$ form a linearly independent set.

3. If two or more eigenvalues are equal it may or may not be possible to get linearly independent eigenvectors corresponding to the equal roots.

4. Two eigenvectors X_1 and X_2 are called orthogonal vectors if $X_1' X_2 = 0$.

5. Eigenvectors of a symmetric matrix corresponding to different eigenvalues are orthogonal.

Normalised form of vectors. To find normalised form of $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$, we divide each element by $\sqrt{a^2+b^2+c^2}$.

For example, normalised form of $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$ is $\begin{bmatrix} 1/3 \\ 2/3 \\ 2/3 \end{bmatrix}$

4.29 NON SYMMETRIC MATRICES WITH NON REPEATED EIGEN VALUES

Example 76. *Find the eigenvalues and eigenvectors of the matrix*

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}.$$

Solution. $|A - \lambda I| = 0$

$$\begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix} = 0 \quad i.e., \quad \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

Let $\lambda = 1$, $\quad 1 - 6 + 11 - 6 = 0$

By synthetic division

$$\begin{array}{c|cccc} 1 & 1 & -6 & +11 & -6 \\ & & -1 & -5 & 6 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

$$\lambda^2 - 5\lambda + 6 = 0$$

$$(\lambda - 1)(\lambda^2 - 5\lambda + 6) = 0$$

$$(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0 \Rightarrow \lambda = 1, 2, 3$$

To find eigenvectors for the corresponding eigenvalues we will consider the matrix equation

$$(A - \lambda I) X = 0$$

$$\begin{bmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \ldots (1)$$

Eigen vector corresponding to eigenvalue $\lambda = 1$

By putting $\lambda = 1$, the matrix equation (1) will become

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \left\{ \begin{aligned} -z &= 0 \\ x + y + z &= 0 \\ k + y + 0 &= 0 \end{aligned} \right\} \quad \begin{aligned} &\text{Let } x = k \\ &\Rightarrow y = -k \end{aligned}$$

Eigen vector X_1 is $\begin{bmatrix} k \\ -k \\ 0 \end{bmatrix}$ or $k\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$

Eigen vector corresponding to eigenvalue $\lambda = 2$.

On putting $\lambda = 2$ in eq. (1). it will become

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -x + 0\,y - z = 0 \\ 2x + 2y + z = 0 \end{matrix}$$

$$\frac{x}{0+2} = \frac{y}{-2+1} = \frac{z}{-2-0} \text{ or } \frac{x}{2} = \frac{y}{-1} = \frac{z}{-2}$$

Eigenvector $X_2 = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = 3$.

On putting $\lambda = 3$ in (1), the equation will become

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} -2x + 0y - z = 0 \\ x - y + z = 0 \end{matrix} \Rightarrow \frac{x}{0-1} = \frac{y}{-1+2} + \frac{z}{2-0}$$

or

$$\frac{x}{1} = \frac{y}{-1} = \frac{z}{-2}$$

Eigenvector $X_3 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$. **Ans.**

Exercise 4.12

Non-symmetric matrix with different eigen values

Find the eigenvalues and the corresponding eigen vectors for the following matrices

1. $\begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ **Ans.** $-1, 1, 2$, $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

2. $\begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$ **Ans.** $1, 2, 5$; $\begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

3. $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ **Ans.** $-2, 1, 3$; $\begin{bmatrix} 11 \\ 1 \\ -14 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

4. $\begin{bmatrix} -9 & 2 & 6 \\ 5 & 0 & -3 \\ -16 & 4 & 11 \end{bmatrix}$ **Ans.** $-1, 1, 2$; $\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$

5. $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ *(A.M.I.E.T.E., Winter 1996)* **Ans.** $-1, 1, 4$; $\begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix}$

6. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 3 & 2 & 3 \end{bmatrix}$ **Ans.** $0, 1, 5$; $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 11 \end{bmatrix}$

7. $\begin{bmatrix} -2 & 1 & 1 \\ -11 & 4 & 5 \\ -1 & 1 & 0 \end{bmatrix}$ **Ans.** $-1, 1, 2;$ $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

4.30 NON-SYMMETRIC MATRIX WITH REPEATED EIGEN VALUES

Example 77. *Find all the Eigenvalues and Eigenvectors of the matrix*

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$ *(A.M.I.E.T.E., Winter 1996)*

Solution. Characteristic equation of A is

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix} = 0$$

or $$(-2-\lambda)[-\lambda^2+\lambda-12]-2(-2\lambda-6)-3(-4+1-\lambda) = 0$$

or $$\lambda^3+\lambda^2-21\lambda-45 = 0 \quad \ldots(1)$$

If $\lambda = -3$, $-27+9+63-45 = 0$, so $(\lambda+3)$ is one factor of (1). The remaining factors are obtained on dividing (1) by $\lambda+3$.

$$\begin{array}{r|rrrr} -3 & 1 & 1 & -21 & -45 \\ & & -3 & 6 & 45 \\ \hline & 1 & -2 & -15 & 0 \end{array}$$

$$\lambda^2-2\lambda-15 = 0 \; or \; (\lambda-5)(\lambda+3) = 0$$

$$(\lambda+3)(\lambda+3)(\lambda-5) = 0 \qquad \Rightarrow \lambda = 5, -3, -3$$

To find the eigenvectors for corresponding eigenvalues, we will consider the matrix equation

$$(A-\lambda I)X = 0 \; i.e., \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \ldots(2)$$

On putting $\lambda = 5$ in eq. (2), it becomes

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We have $-7x+2y-3z = 0$, $2x-4y-6z = 0$, $-x-2y-5z = 0$

From first and second equations, we have

$$\frac{x}{-12-12} = \frac{y}{-6-42} = \frac{z}{28-4} \text{ or } \frac{x}{-24} = \frac{y}{-48} = \frac{z}{24} \text{ or } \frac{x}{1} = \frac{y}{2} = \frac{z}{-1} = k$$

Hence the eigenvector is $X_1 = \begin{bmatrix} k \\ 2k \\ -k \end{bmatrix}$ or $\begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

Put $\lambda = -3$ in eq. (2)

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We have $x+2y-3z = 0$, $2x+4y-6z = 0$, $-x-2y+3z = 0$

First, second and third equations are the same.

Let $x = k_1$, $y = k_2$ then $z = \frac{1}{3}(k_1 + 2k_2)$

Hence, the eigenvector is $\begin{bmatrix} k_1 \\ k_2 \\ \frac{1}{3}(k_1 + 2k_2) \end{bmatrix}$

Let $k_1 = 0$, $k_2 = 3$, Hence $X_2 = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}$

Since the matrix is non-symmetric the corresponding eigenvectors X_2 and X_3 must be linearly independent. This can be done by choosing

$k_2 = 0$, and $k_1 = 1$, Hence $X_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$

$$\lambda_1 = 5 \rightarrow X_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \ \lambda = -3 \rightarrow X_2 = \begin{bmatrix} 0 \\ 3 \\ 2 \end{bmatrix}, \ \lambda = -3 \rightarrow X_3 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}. \quad \textbf{Ans.}$$

Example 78. *Show that the matrix* $A \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ *has less than three linearly independent eigenvectors. Is it possible to obtain a similarity transformation that will diagonalise this matrix?* (*A.M.I.E.T.E., Summer 1996*)

Solution. $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 10 & 5 \\ -2 & -3-\lambda & -4 \\ 3 & 5 & 7-\lambda \end{vmatrix} = 0 \text{ or } \lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0$$

Let $\lambda = 2$, $8 - 28 + 32 - 12 = 0$. $\therefore (\lambda - 2)$ is one factor.

$$\begin{array}{r|rrrr} 2 & 1 & -7 & 16 & -12 \\ & & 2 & -10 & 12 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

$$\lambda^2 - 5 + 6 = 0$$

$$(\lambda - 2)(\lambda^2 - 5\lambda + 6) = 0 \text{ or } (\lambda - 2)(\lambda - 2)(\lambda - 3) = 0, \ \lambda = 2, 2, 3$$

Eigenvector for $\lambda = 3$

$$\begin{bmatrix} 3-3 & 10 & 5 \\ -2 & -3-3 & -4 \\ 3 & 5 & 7-3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 10 & 5 \\ -2 & -6 & -4 \\ 3 & 5 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\left.\begin{aligned} 0x + 10y + 5z &= 0 \\ -2x - 6y - 4z &= 0 \end{aligned}\right] \Rightarrow \frac{x}{-40+30} = \frac{y}{-10-0} = \frac{z}{0+20} \text{ or } \frac{x}{1} = \frac{y}{1} = \frac{z}{-2}$$

Eigenvector is $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

Eigenvector for $\lambda = 2$

$$\begin{bmatrix} 1 & 10 & 5 \\ -2 & -5 & -4 \\ 3 & 5 & 5 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \ i.e., \left.\begin{aligned} x + 10y + 5z &= 0 \\ 2x + 5y + 4z &= 0 \end{aligned}\right] \Rightarrow \frac{x}{40-25} = \frac{y}{10-4} = \frac{z}{5-20}$$

or $\frac{x}{5} = \frac{y}{2} = \frac{z}{-5} = k$, Eigenvector $= \begin{bmatrix} 5k \\ 2k \\ -5k \end{bmatrix}$

We get one eigenvector corresponding to repeated root $\lambda_2 = 2 = \lambda_3$.

Eigenvectors corresponding to $\lambda_2 = 2 = \lambda_3$ are not linearly independent.

Similarity transformation is not possible.

Example 79. *Find the eigenvalues and the corresponding eigenvectors for the matrix*

$$A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$$

Solution.
$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{vmatrix} = 0 \text{ or } \lambda^3 - 2\lambda^2 + \lambda = 0$$

$$\lambda(\lambda^2 - 2\lambda + 1) = 0$$

$$\lambda(\lambda - 1)^2 = 0$$

$\lambda = 0, 1, 1$

Matrix equation for eigenvector $[A - \lambda I]X = 0$

$$\begin{bmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \ldots (1)$$

Eigen vector corresponding to eigenvalue $\lambda = 0$.

On putting $\lambda = 0$ in (1), the equation will become

$$\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} x - 6y - 4z = 0 \\ 0x + 4y + 2z = 0 \end{bmatrix}$$

$$\frac{x}{-12+16} = \frac{y}{0-2} = \frac{z}{4-0} \text{ or } \frac{x}{2} = \frac{y}{-1} = \frac{z}{2}$$

Eigenvector $X_1 = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = 1$.

On putting $\lambda = 1$ in (1), the equation will become

$$\begin{bmatrix} 0 & -6 & -4 \\ 0 & 3 & 2 \\ 0 & -6 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow -6y - 4z = 0 \text{ or } y = -\frac{4}{6}z \quad (z = 3)$$

x can take any value say $x = 1$, $y = -2$, $z = 3$

Eigenvector $X_2 = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

The corresponding eigenvector X_2 and X_3 must be linearly independent.

$$X_3 = \begin{bmatrix} 0 \\ 2 \\ -3 \end{bmatrix}$$

To make X_3 linearly independent we have chosen $x = 0$) **Ans.**

Exercise 4.13

Non-symmetric matrices with repeated eigenvalues

Find the eigenvalues and eigenvectors of the following matrices :

1. $\begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ **Ans.** $-2, 2, 2; \begin{bmatrix} -4 \\ -1 \\ +7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

2. $\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ (*A.M.I.E.T.E., Winter 2003, Summer 1997*) **Ans.** $1, 1, 5; k\begin{bmatrix} 1 \\ 2 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

3. $\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$ **Ans.** $1, 1, 7; \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

4. $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ **Ans.** $-3, -3, 5; \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$

5. $\begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ **Ans.** $-1, -1, 3; \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$

6. $\begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ (*A.M.I.E.T.E., Summer 1996*) **Ans.** $2, 2, 3; \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ -2 \\ 5 \end{bmatrix}$ Dependent Eigen vectors

4.31 SYMMETRIC MATRICES WITH NON REPEATED EIGENVALUES

Example 80. *Find the eigenvalues and the corresponding eigenvectors of the matrix*

$$\begin{bmatrix} -2 & 5 & 4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{bmatrix}$$

Solution. $|A - \lambda I| = 0$

$$\begin{vmatrix} -2-\lambda & 5 & 4 \\ 5 & 7-\lambda & 5 \\ 4 & 5 & -2-\lambda \end{vmatrix} = 0 \quad \text{or} \quad \lambda^3 - 3\lambda^2 - 90\lambda - 216 = 0$$

Take $\lambda = -3$, $\quad -27 - 27 + 270 - 216 = 0$

By synthetic division

$$\begin{array}{rrrrr} -3 & 1 & -3 & -90 & -216 \\ & & -3 & 18 & 216 \\ \hline & 1 & -6 & -72 & 0 \end{array}$$

$$\lambda^2 - 6\lambda - 72 = 0$$

$$(\lambda - 12)(\lambda + 6) = 0$$

$$\lambda = -3, -6, 12$$

Matrix equation for eigenvectors $[A - \lambda I]X = 0$

$$\begin{bmatrix} -2-\lambda & 5 & 4 \\ 5 & 7-\lambda & 5 \\ 4 & 5 & -2-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \dots (1)$$

On putting $\lambda = -3$ in eq. (1), it will become

$$\begin{bmatrix} 1 & 5 & 4 \\ 5 & 10 & 5 \\ 4 & 5 & 1 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} x + 5y + 4z = 0 \\ 5x + 10y + 5z = 0 \end{cases}$$

$$\frac{x}{25-40} = \frac{y}{20-5} = \frac{z}{10-25} \text{ or } \frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$$

Eigenvector $X_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$.

Eigenvector corresponding to eigenvalue $\lambda = -6$.

Equaton (1) becomes

$$\begin{bmatrix} 4 & 5 & 4 \\ 5 & 13 & 5 \\ 4 & 5 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{matrix} 4x + 5y + 4z = 0 \\ 5x + 13y + 5z = 0 \end{matrix}$$

$$\frac{x}{25-52} = \frac{y}{20-20} = \frac{z}{52-25} \text{ or } \frac{x}{1} = \frac{y}{0} = \frac{z}{-1}$$

Eigenvector $X_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

Eigenvector corresponding to Eigenvalue 12.

Equation (1) becomes

$$\begin{bmatrix} -14 & 5 & 4 \\ 5 & -5 & 5 \\ 4 & 5 & -14 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{cases} -14x + 5y + 4z = 0 \\ 5x - 5y + 5z = 0 \end{cases}$$

$$\frac{x}{25+20} = \frac{y}{20+70} = \frac{z}{70-25} \text{ or } \frac{x}{1} = \frac{y}{2} = \frac{z}{1}$$

Eigenvector $X_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$ **Ans.**

Exercise 4.14

Symmetric matrices with non-repeated eigenvalues

Find the eigen values and eigenvectors of the following matrices:

1. $\begin{bmatrix} 5 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 5 \end{bmatrix}$ **Ans.** $-2, 4, 6$; $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

2. $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ **Ans.** 2, 3, 6; $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$

3. $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ *(A.M.I.E.T.E., Winter 2002)* **Ans.** 0, 3, 15; $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

4. $\begin{bmatrix} 2 & 4 & -6 \\ 4 & 2 & -6 \\ -6 & -6 & -15 \end{bmatrix}$ **Ans.** – 2, 9, – 18; $\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix}$

5. $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ **Ans.** –2, 3, 6 $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

4.32 SYMMETRIC MATRICES WITH REPEATED EIGENVALUES

Example 81. *Find all the eigenvalues and eigenvectors of the matrix*

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Soluton. The characteristic equation is

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

or $(2-\lambda)[(2-\lambda)^2-1]+1[-2+\lambda+1]+1[1-2+\lambda] = 0$

or $(2-\lambda)(4-4\lambda+\lambda^2-1)+(\lambda-1)+\lambda-1 = 0$

or $8-8\lambda+2\lambda^2-2-4\lambda+4\lambda^2-\lambda^3+\lambda+2\lambda-2 = 0$

or $-\lambda^3+6\lambda^2-9\lambda+4 = 0$

or $\lambda^3-6\lambda^2+9\lambda-4 = 0$ ...(1)

On putting $\lambda = 1$ in (1), the equation (1) is satisfied, so $\lambda - 1$ is one factor of the equation (1). The other factor $(\lambda^2 - 5\lambda + 4)$ is got on dividing (1) by $\lambda - 1$.

or $(\lambda-1)(\lambda^2-5\lambda+4) = 0$ or $(\lambda-1)(\lambda-1)(\lambda-4) = 0$

$\therefore$ $\lambda = 1, 1, 4$

The eigenvalues are 1, 1, 4.

When $\lambda = 4$ $\begin{pmatrix} 2-4 & -1 & 1 \\ -1 & 2-4 & -1 \\ 1 & -1 & 2-4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$

or $\begin{pmatrix} -2 & -1 & 1 \\ -1 & -2 & -1 \\ 1 & -1 & -2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$

On interchanging first and third rows

$$\begin{pmatrix} 1 & -1 & -2 \\ -1 & -2 & -1 \\ -2 & -1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$R_2 + R_1,\ R_3 + 2R_1 \qquad\qquad\qquad R_3 + R_2$$

$$\begin{pmatrix} 1 & -1 & -2 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0, \quad \begin{pmatrix} 1 & -1 & -2 \\ 0 & -3 & -3 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

$$x_1 - x_2 - 2x_3 = 0$$

$$-3x_2 - 3x_3 = 0$$

Let $x_3 = k$ $\qquad -3x_2 - 3k = 0 \quad$ or $\quad x_2 = -k$

$\qquad x_1 + k - 2k = 0 \quad$ or $\quad x_1 = k$

Let $k = 1$

The eigenvector $\qquad X_1' = [1, -1, 1]'$

When $\lambda = 1$ $\qquad \begin{pmatrix} 2-1 & -1 & 1 \\ -1 & 2-1 & -1 \\ 1 & -1 & 2-1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$

$$R_2 + R_1 \text{ and } R_3 - R_1$$

or $\begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$ or $\begin{pmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$

$$x_1 - x_2 + x_3 = 0$$

Let $x_1 = k_1$ and $x_2 = k_2$

$$k_1 - k_2 + x_3 = 0 \quad \text{or} \quad x_3 = k_2 - k_1$$

Eigenvector is $[k_1, k_2, k_2 - k_1]'$

Choosing $k_1 = 1$, $k_2 = 1$ then eigenvector is $[1, 1, 0]'$, choosing $k_1 = 1$, $k_2 = 2$ then eigen vector $X_2' = [1, 2, -1]'$

Let $X_3 = \begin{bmatrix} l \\ m \\ n \end{bmatrix}$ as X_3 is orthogonal to X_1 and X_2 since the given matrix is symmetric

$$[1, -1, 1]\begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0 \quad \text{or} \quad l - m + n = 0 \qquad \text{... (2)}$$

$$[1, 1, 0]\begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0 \quad \text{or} \quad l + m + o\,n = 0 \qquad \text{... (3)}$$

Solving (2) and (3), we get $\dfrac{l}{-1} = \dfrac{m}{1} = \dfrac{n}{2}$

$$X_3 = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \qquad \textbf{Ans.}$$

Exercise 4.15

Symmetric matrices with repeated eigenvalues

Find the eigenvalues and the corresponding eigen vectors of the following marices:

1. $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ **Ans.** 0, 0, 14; $\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 6 \\ -5 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

2. $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$ **Ans.** 1, 3, 3; $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$

3. $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ **Ans.** 8, 2, 2; $\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$

4. Choose the correct or the best of the answers given in the following parts:

(*i*) Two of the eigenvalues of a 3×3 matrix, whose determinant equals 4, are, -1 and $+2$ the third eigen value of the matrix is euqal to

(*a*) -2 (*b*) -1 (*c*) 1 (*d*) 2

(*ii*) If a square matrix A has an eigenvalue λ, then an eigenvalue of the matrix $(kA)^T$ where, $k \neq 0$, is a scalar is

(a) λ / k (b) k / λ (c) $k\lambda$ (d) None of these

(*A.M.I.E.T.E., Summer 1996*)

(*iii*) An eigenvalue of a square matrix A is $\lambda = 0$. Then

(*a*) $|A| \neq 0$; (*b*) A is symmetric ; (*c*) A is singular ;
(*d*) A is skew-symmetric ; (*e*) A is an even order matrix : (*f*) A is an odd order matrix.

(*A.M.I.E.T.E., Summer 1996*)

(*iv*) The matrix A is defined as $A = \begin{bmatrix} -1 & 0 & 0 \\ 2 & -3 & 0 \\ 1 & 4 & 2 \end{bmatrix}$. The eigenvalues of A^2 are

(*a*) $-1, -9, -4$, (*b*) 1, 9, 4, (*c*) $-1, -3, 2$, (*d*) $1, 3, -2$.

(*v*) If the matrix is $A = \begin{bmatrix} -1 & 2 & 3 \\ 0 & 3 & 5 \\ 0 & 0 & -2 \end{bmatrix}$ then the eigenvalues of $A^3 + 5A + 8I$, are

(*a*) $-1, 27, -8$; (*b*) $-1, 3, -2$; (*c*) $2, 50, -10$; (*d*) 2, 50, 10.

(*vi*) The matrix A has eigen values $\lambda_i \neq 0$. Then $A^{-1} - 2I + A$ has eigenvalues

(*a*) $1 + 2\lambda_i + \lambda_i^2$ (*b*) $\dfrac{1}{\lambda_i} - 2 + \lambda_i$ (*c*) $1 - 2\lambda_i + \lambda_i^2$ (*d*) $1 - \dfrac{2}{\lambda_i} + \dfrac{1}{\lambda_i^2}$

(*vii*) The eigen values of a matrix A are $1, -2, 3$. The eigenvalues of $3I - 2A + A^2$ arę

(*a*) 2, 11, 6 (*b*) 3, 11, 18 (*c*) 2, 3, 6 (*d*) 6, 3, 11.

Ans. (*i*) (*b*), (*ii*) (*c*), (*iii*) (*c*), (*iv*) (*b*), (*v*) (*c*), (*vi*) (*b*), (*vii*) (*a*)

4.33 DIAGONALISATION OF A MATRIX

Theorem. If a square matrix A of order n has n linearly independent eigenvectors, then a matrix P can be found such that $P^{-1}AP$ is a diagonal matrix.

Proof. We shall prove the theorem for a matrix of order 3. The proof can be easily extended to matrices of higher order.

Let
$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

and let $\lambda_1, \lambda_2, \lambda_3$ be its eigen values and X_1, X_2, X_3 the corresponding eigen vectors, where

$$X_1 = \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix},\ X_2 = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix},\ X_3 = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix}$$

For the eigenvalues λ_1, the eigenvector is given by

$$\left.\begin{aligned} (a_1-\lambda_1)\,x_1 + b_1y_1 + c_1\,z_1 &= 0 \\ a_2\,x_1 + (b_2-\lambda_1)\,y_1 + c_2\,z_1 &= 0 \\ a_3\,x_1 + b_3y_1 + (c_3-\lambda_1)\,z_1 &= 0 \end{aligned}\right\} \quad \ldots (i)$$

$\therefore$ We have

$$\left.\begin{aligned} a_1x_1 + b_1y_1 + c_1z_1 &= \lambda_1x_1 \\ a_2x_1 + b_2y_1 + c_2z_1 &= \lambda_1y_1 \\ a_3x_1 + b_3y_1 + c_3z_1 &= \lambda_1z_1 \end{aligned}\right\} \quad \ldots (ii)$$

Similarly for λ_2 and λ_3 we have

$$\left.\begin{aligned} a_1x_2 + b_1y_2 + c_1z_2 &= \lambda_2x_2 \\ a_2x_2 + b_2y_2 + c_2z_2 &= \lambda_2y_2 \\ a_3x_2 + b_3y_2 + c_3z_2 &= \lambda_2z_2 \end{aligned}\right\} \quad \ldots (iii)$$

and

$$\left.\begin{aligned} a_1x_3 + b_1y_3 + c_1z_3 &= \lambda_3x_3 \\ a_2x_3 + b_2y_3 + c_2z_3 &= \lambda_3y_3 \\ a_3x_3 + b_3y_3 + c_3z_3 &= \lambda_3z_3 \end{aligned}\right\} \quad \ldots (iv)$$

We consider the matrix
$$P = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

whose columns are the eigenvectors of A.

Then
$$AP = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix}$$

$$= \begin{pmatrix} a_1x_1 + b_1y_1 + c_1z_1 & a_1x_2 + b_1y_2 + c_1z_2 & a_1x_3 + b_1y_3 + c_1z_3 \\ a_2x_1 + b_2y_1 + c_2z_1 & a_2x_2 + b_2y_2 + c_2z_2 & a_2x_3 + b_2y_3 + c_2z_3 \\ a_3x_1 + b_3y_1 + c_3z_1 & a_3x_2 + b_3y_2 + c_3z_2 & a_3x_3 + b_3y_3 + c_3z_3 \end{pmatrix}$$

$$= \begin{pmatrix} \lambda_1x_1 & \lambda_2x_2 & \lambda_3x_3 \\ \lambda_1y_1 & \lambda_2y_2 & \lambda_3y_3 \\ \lambda_1z_1 & \lambda_2z_2 & \lambda_3z_3 \end{pmatrix} \quad \text{[using results } (ii), (iii) \text{ and } (iv)$$

$$= \begin{pmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = PD$$

where D is the Diagonal matrix $\begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$

$$\therefore \qquad AP = PD$$

or $$P^{-1}AP = P^{-1}PD = D$$

Note 1. The square matrix P, which diagonalises A, is found by grouping the eigen vectors of A into square-matrix and the resulting diagonal matrix has the eigen values of A as its diagonal elements.

2. The transformation of a matrix A to $P^{-1}AP$ is known as a *similarity transformation.*

3. The reduction of A to a diagonal matrix is, obviously, a particular case of similarity transformation.

4. The matrix P which diagonalises A is called the modal matrix of A and the resulting diagonal matrix D is known as the spectra matrix of A.

Example 82. *A square matrix A is defined by* $A = \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix}$. *Find the modal matrix P and the resulting diagonal matrix D* of A. (*A.M.I.E.T.E., Summer 2001, Summer 1999*)

Solution. The characteristic equation of the matrix A is

$$\begin{vmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{vmatrix} = 0$$

or $$(-1-\lambda)(-2\lambda+\lambda^2+1) - 2(-\lambda+1) - 2(-1+2-\lambda) = 0$$

or $$(-\lambda-1)(\lambda^2-2\lambda+1) + 2\lambda - 2 - 2 + 2\lambda = 0$$

or $$(-\lambda-1)(\lambda-1)^2 + (4\lambda-4) = 0$$

or $$(\lambda-1)(\lambda^2-5) = 0, \ \lambda = 1, \ \lambda = \pm\sqrt{5}$$

Eigenvalues are $1, -\sqrt{5}, \sqrt{5}$.

Eigenvectors are given by the matrix equation

$$\begin{bmatrix} -1-\lambda & 2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & 0-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

for $\lambda = 1$, $$\begin{bmatrix} -2 & 2 & -2 \\ 1 & 1 & 1 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We have $$-2x + 2y - 2z = 0$$
$$x + y + z = 0$$

From first and second equations, we have

$$\frac{x}{2+2} = \frac{y}{-2+2} = \frac{z}{-2-2} \text{ or } \frac{x}{4} = \frac{y}{0} = \frac{z}{-4} = k$$

Hence the eigenvector $X_1 = \begin{bmatrix} 4k \\ 0 \\ -4k \end{bmatrix}$ or $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

When $\lambda = -\sqrt{5}$, $\begin{bmatrix} -1+\sqrt{5} & 2 & -2 \\ 1 & 2+\sqrt{5} & 1 \\ -1 & -1 & 0+\sqrt{5} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

We have $(-1+\sqrt{5})\,x + 2y - 2z = 0,\ x + (2+\sqrt{5})\,y + z = 0,\ -x - y + \sqrt{5}\,z = 0$

From first and second equations we have

$$\frac{x}{6+2\sqrt{5}} = \frac{y}{-1-\sqrt{5}} = \frac{z}{1+\sqrt{5}} \text{ or } \frac{x}{1+\sqrt{5}} = \frac{y}{-1} = \frac{z}{1}$$

Hence the eigenvector is $X_2 = \begin{bmatrix} 1+\sqrt{5} \\ -1 \\ 1 \end{bmatrix}$.

When $\lambda = \sqrt{5}$, $\begin{bmatrix} -1-\sqrt{5} & 2 & -2 \\ 1 & 2-\sqrt{5} & 1 \\ -1 & -1 & 0-\sqrt{5} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

We have $(-1-\sqrt{5})\,x + 2y - 2z = 0,\ x + (2-\sqrt{5})\,y + z = 0,\ -x - y - \sqrt{5}\,z = 0$

From first and second equations we have

$$\frac{x}{6-2\sqrt{5}} = \frac{y}{-1+\sqrt{5}} = \frac{z}{1-\sqrt{5}} \text{ or } \frac{x}{1-\sqrt{5}} = \frac{y}{-1} = \frac{z}{1}$$

Hence the eigenvector $X_3 = \begin{bmatrix} 1-\sqrt{5} \\ -1 \\ 1 \end{bmatrix}$.

Modal Matrix $P = \begin{bmatrix} -1 & 1+\sqrt{5} & 1-\sqrt{5} \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$

$$P^{-1}AP = \frac{1}{2\sqrt{5}} \begin{bmatrix} 0 & 2\sqrt{5} & 2\sqrt{5} \\ -1 & -2-\sqrt{5} & -1 \\ 1 & 2-\sqrt{5} & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & -2 \\ 1 & 2 & 1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 1-\sqrt{5} & 1+\sqrt{5} \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2\sqrt{5}} \begin{bmatrix} 0 & 2\sqrt{5} & 2\sqrt{5} \\ -\sqrt{5} & -5-2\sqrt{5} & -\sqrt{5} \\ -\sqrt{5} & 5-2\sqrt{5} & -\sqrt{5} \end{bmatrix} \begin{bmatrix} -1 & 1-\sqrt{5} & 1+\sqrt{5} \\ 0 & -1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{2\sqrt{5}} \begin{bmatrix} 2\sqrt{5} & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & -10 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sqrt{5} & 0 \\ 0 & 0 & -\sqrt{5} \end{bmatrix} = D.$$

Example 83. *Let* $A = \begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$

Find a similarity transformation that diagonalises matrix A. *(Allahabad 1997)*

Solution. $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

Characteristic Eqn. is $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} -(2+\lambda) & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$\Rightarrow -(2+\lambda)[(1-\lambda)(-\lambda)-12]-2[-2\lambda-6]-3[-4+(1-\lambda)]=0$

$\Rightarrow -(2+\lambda)[-\lambda+\lambda^2-12]+4\lambda+12+12-3+3\lambda = 0$

$\Rightarrow 2\lambda-2\lambda^2+24+\lambda^2-\lambda^3+12\lambda+4\lambda+12+12-3+3\lambda = 0$

$\Rightarrow -\lambda^3-\lambda^2+21\lambda+45 = 0 \Rightarrow \lambda^3+\lambda^2-21\lambda-45 = 0$

$\Rightarrow (\lambda-5)(\lambda^2+6\lambda+9) = 0, \Rightarrow (\lambda-5)(\lambda+3)^2 = 0$

$\Rightarrow \lambda = 5, -3, -3.$

Eigenvector for $\lambda = 5$ is

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} -1 & -2 & -5 \\ 2 & -4 & -6 \\ -7 & 2 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -2 & -5 \\ 0 & -8 & -16 \\ 0 & 16 & 32 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} R_2+2R_1 \\ \\ R_3-7R_1 \end{matrix}$$

$$\begin{bmatrix} -1 & -2 & -5 \\ 0 & -8 & -16 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \; R_3+2R_2$$

$$-x-2y-5z = 0$$

$$-8y-16z = 0$$

Let $z = 1 \therefore \quad -8y-16 = 0 \text{ or } y = -2$

$\quad -x+4-5 = 0 \text{ or } x = -1$

Eigenvector is $(-1, -2, 1)'$

For $\lambda = -3 \quad \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \begin{matrix} R_2-2R_1 \\ R_3+R_1 \end{matrix}$$

$$x+2y-3z = 0$$

Let $\quad z = 0, y = -1, x-2+0 = 0 \text{ or } x = 2$

Eigenvector is $[2 \;\; -1 \;\; 0]'$

Let $\quad z = 1, y = 0, x+0-3 = 0 \text{ or } x = 3$

Eigenvector is $(3, \;\; 0, \;\; 1)'$

We have got three eigen vectors *i.e.*,

$$(-1, -2, 1)', (2, -1, 0)', (3, 0, 1)'$$

Three independent eigen vectors, writing as three column vectors, we get the required transformation matrix.

$$B = \begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \text{ and } B^{-1} = \frac{1}{8}\begin{bmatrix} -1 & -2 & 3 \\ 2 & -4 & -6 \\ 1 & 2 & 5 \end{bmatrix}$$

$$B^{-1}AB = \frac{1}{8}\begin{bmatrix} -1 & -2 & 3 \\ 2 & -4 & -6 \\ 1 & 2 & 5 \end{bmatrix}\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 6 \end{bmatrix}\begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix} -5 & -10 & 15 \\ -6 & 12 & 18 \\ -3 & -6 & -15 \end{bmatrix}\begin{bmatrix} -1 & 2 & 3 \\ -2 & -1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \frac{1}{8}\begin{bmatrix} 40 & 0 & 0 \\ 0 & -24 & 0 \\ 0 & 0 & -24 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -3 \end{bmatrix} \quad \textbf{Ans.}$$

Example 84. *Let* $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Find matrix P such that $P^{-1}AP$ *is diagonal matrix.*

Solution. The characteristic equation of the matrix A is

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

or $\quad (6-\lambda)[9+\lambda^2-6\lambda-1]+2[-6+2\lambda+2]+2[2-6+2\lambda] = 0$

or $\quad (6-\lambda)(\lambda^2-6\lambda+8)-8+4\lambda-8+4\lambda = 0$

or $\quad 6\lambda^2-36\lambda+48-\lambda^3+6\lambda^2-8\lambda-16+8\lambda = 0$

or $\quad -\lambda^3+12\lambda^2-36\lambda+32 = 0 \quad$ or $\quad \lambda^3-12\lambda^2+36\lambda-32 = 0$

or $\quad (\lambda-2)^2(\lambda-8) = 0$ or $\lambda = 2, 2, 8$

Eigen vector for $\lambda = 2$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 2 & -1 & 1 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \frac{1}{2}R_1$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{or} \quad 2x_1 - x_2 + x_3 = 0$$

This equation is satisfied by $x_1 = 0, x_2 = 1, x_3 = 1$ and $x_1 = 1, x_2 = 3, x_3 = 1$.

Eigen vectors are $[0, 1, 1]', [1, 3, 1]'$.

Eigenvector for $\lambda = 8$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

or
$$\begin{bmatrix} 1 & 1 & -1 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \ \frac{1}{2}R_1$$

$R_2 + 2R_1, R_3 - 2R_1$

or
$$\begin{bmatrix} 1 & 1 & -1 \\ 0 & -3 & -3 \\ 0 & -3 & -3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ or } \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + x_2 - x_3 = 0 \text{ and } x_2 + x_3 = 0$$

Let $x_3 = 1, x_2 = -1$ and so $x_1 = 2$.

Eigenvector is $[2, -1, 1]'$

$$\therefore \quad P = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}, \ P^{-1} = -\frac{1}{6}\begin{bmatrix} 4 & 1 & -7 \\ -2 & -2 & 2 \\ -2 & 1 & -1 \end{bmatrix}$$

Now
$$P^{-1}AP = \frac{1}{6}\begin{bmatrix} 4 & 1 & -7 \\ -2 & -2 & 2 \\ -2 & 1 & -1 \end{bmatrix}\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}\begin{bmatrix} 0 & 1 & 2 \\ 1 & 3 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 8 \end{bmatrix} \quad \textbf{Ans.}$$

Example 85. *The matrix* $A = \begin{bmatrix} a & h \\ h & b \end{bmatrix}$ *is transformed to the diagonal form* $D = T^{-1}AT$, *where* $T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$. *Find the value of* θ *which gives this diagonal transformation.*

Solution.
$$T = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \therefore T^{-1} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Now
$$T^{-1}AT = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} a & h \\ h & b \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} a\cos\theta - h\sin\theta & h\cos\theta - b\sin\theta \\ a\sin\theta + h\cos\theta & h\sin\theta + b\cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} a\cos^2\theta - 2h\sin\theta\cos\theta + b\sin^2\theta & (a-b)\sin\theta\cos\theta - h\sin^2\theta + h\cos^2\theta \\ (a-b)\sin\theta\cos\theta + h\cos^2\theta - h\sin^2\theta & a\sin^2\theta + 2h\sin\theta\cos\theta + b\cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} a\cos^2\theta - h\sin 2\theta + b\sin^2\theta & (a-b)\sin\theta\cos\theta + h\cos 2\theta \\ (a-b)\sin\theta\cos\theta + h\cos 2\theta & a\sin^2\theta + h\sin 2\theta + b\cos^2\theta \end{bmatrix}$$

$$= \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \quad \text{Being diagonal matrix}$$

$$\therefore \quad (a-b)\sin\theta\cos\theta + h\cos 2\theta = 0$$

or $\qquad \tan 2\theta = \dfrac{2h}{b-a} \quad$ or $\quad \theta = \dfrac{1}{2}\tan^{-1}\dfrac{2h}{b-a} \qquad$ **Ans.**

Exercise 4.16

1. Find the modal matri of the following matrices

$$(i) = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix} \quad (ii) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \quad \textbf{Ans. } (i) = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \quad (ii) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

(A.M.I.E.T.E., winter 2000) *(R.G.V.P., Bhopal, I Sem.)*

2. For the matrix $A = \begin{bmatrix} 4 & 1 & 0 \\ 1 & 4 & 1 \\ 0 & 1 & 4 \end{bmatrix}$, determine a matrix P such that $P^{-1}AP$ is diagonal matrix.

(A.M.I.E.T.E., Summer 1997) $\quad P = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -\sqrt{2} & \sqrt{2} \\ 1 & 1 & 1 \end{bmatrix}$

3. Determine the eigen values and the corresponding egienvectors of the matrix

$$A = \begin{bmatrix} 5 & 7 & -5 \\ 0 & 4 & -1 \\ 2 & 8 & -3 \end{bmatrix}$$

Hence find the matrix B such that $B^{-1}AB$ is diagonal matrix. *(A.M.I.E.T.E., Winter 1997)*

$$\textbf{Ans. } B = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 1 & 1 \\ 3 & 2 & 1 \end{bmatrix}$$

4. Diagonalise the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ *(A.M.I.E.T.E., Summer 2003)* $\quad$ **Ans.** $\begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

5. Reduce the following matrix A into a diagonal matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$ *(A.M.I.E.T.E., Summer 2000)* $\quad$ **Ans.** $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$

4.34 POWERS OF A MATRIX

We can obtain powers of a matrix by using diagonalisation.

We know that $D = P^{-1}AP$

Where A is the square matrix and P is a non-singular matrix.

$$D^2 = (P^{-1}AP)(P^{-1}AP) = P^{-1}A(PP^{-1})AP = P^{-1}A^2P$$

Similarly $D^3 = P^{-1}A^3P$

In general $D^n = P^{-1}A^nP \qquad \ldots (1)$

Pre-multiply (1) by P and post-multiply by P^{-1}

$$PD^nP^{-1} = P(P^{-1}A^nP)P^{-1}$$
$$= (PP^{-1})A^n(PP^{-1})$$
$$= A^n$$

Procedure: (1) Find eigenvalues for a square matrix A.